PEER-REVIEWED JOURNAL OF COMPUTER SCIENCE (PRJCS)

ISSN Online :

Open Access Scholarly Journal

Federated Learning for Privacy-Preserving Healthcare Data Analytics

Manasy Jayasurya

Assistant Professor, Department of Computer Science and Applications, St. Mary's College (Autonomous), Thrissur, India

Article information

Received: 12th February 2026
Received in revised form: 25th February 2026
Accepted: 1st March 2026
Available online: 9th March 2026

Volume: 1
Issue: 3
DOI: https://doi.org/10.5281/zenodo.19247308

Abstract

The proliferation of electronic health records (EHRs), medical imaging repositories, and wearable sensor data has created unprecedented opportunities for data-driven healthcare analytics. However, stringent privacy regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) fundamentally constrain the centralized aggregation of sensitive patient data across institutional boundaries. Federated learning (FL) has emerged as a transformative paradigm that enables collaborative model training across distributed healthcare institutions without requiring raw data exchange. This paper presents a comprehensive survey of federated learning methodologies applied to privacy-preserving healthcare data analytics. We systematically examine the architectural foundations of FL, including horizontal and vertical federated learning, federated transfer learning, and split learning variants. We provide a detailed taxonomy of privacy-enhancing mechanisms integrated with FL frameworks, encompassing differential privacy, secure multi-party computation, homomorphic encryption, and trusted execution environments. The paper critically evaluates FL applications across diverse healthcare domains, including medical image analysis, electronic health record mining, genomics, drug discovery, and remote patient monitoring. Furthermore, we analyze key challenges related to statistical heterogeneity, communication efficiency, system heterogeneity, and adversarial robustness in healthcare FL deployments. Our analysis demonstrates that federated learning achieves performance within 1.5–3.2% of centralized baselines across multiple healthcare tasks while maintaining rigorous privacy guarantees. We conclude by identifying open research directions and proposing a roadmap for the clinical deployment of federated learning systems.

Keywords: - Federated Learning, Privacy-Preserving Machine Learning, Healthcare Analytics, Differential Privacy, Electronic Health Records, Medical Imaging, Distributed Machine Learning

I. INTRODUCTION

The digital transformation of healthcare systems has generated vast quantities of clinical data distributed across hospitals, research institutions, and primary care networks worldwide. Electronic health records alone encompass over 2.3 exabytes of data globally, with projections indicating a compound annual growth rate of 36% through 2030 [1]. This data deluge presents immense potential for machine learning applications in disease prediction, treatment optimization, and population health management. However, the inherently sensitive nature of health data imposes fundamental constraints on conventional centralized machine learning approaches that require data aggregation at a single site [2].

Privacy regulations across jurisdictions—including HIPAA in the United States, GDPR in the European Union, and PIPEDA in Canada—establish strict limitations on the transfer and processing of protected health information (PHI). These regulatory frameworks, while essential for patient privacy protection, create data silos that impede the development of robust, generalizable machine learning models for healthcare applications [3]. The resulting fragmentation of healthcare data across institutional boundaries represents one of the most significant barriers to realizing the full potential of artificial intelligence in medicine [4].

Federated learning (FL), first proposed by McMahan et al. [5] in the context of mobile keyboard prediction, offers a paradigm shift by enabling collaborative model training without centralizing raw data. In the FL framework, a global model is iteratively improved through rounds of local training at participating institutions followed by aggregation of model updates (typically gradients or weights) at a central server. This approach preserves data locality while enabling institutions to collectively benefit from larger effective training datasets [6]. The federated learning process, illustrating the interplay between local training rounds and global aggregation across communication rounds, is depicted in Figure 1.

The healthcare domain is particularly well-suited for federated learning due to several factors:

- patient data is naturally distributed across multiple hospitals and clinics
- data sharing restrictions are legally mandated
- rare disease datasets at individual institutions are often insufficient for training robust models
- healthcare institutions have strong incentives to collaborate for improved patient outcomes [7].

Recent landmark studies have demonstrated the feasibility of FL in multi-institutional medical imaging [8], electronic health record analysis [9], and genomic studies [10].

This paper provides a comprehensive survey of federated learning for privacy-preserving healthcare data analytics. Our contributions are fourfold:

- we present a systematic taxonomy of FL architectures and their applicability to healthcare settings
- we provide a detailed analysis of privacy mechanisms integrated with FL, as summarized in Table 1
- we critically review FL applications across major healthcare domains with performance benchmarks presented in Table 2 and Figure 2
- we identify open challenges and propose research directions for clinical FL deployment.

II. FOUNDATIONS OF FEDERATED LEARNING

A. Federated Averaging and Core Mechanisms

The foundational algorithm for federated learning, Federated Averaging (FedAvg), was introduced by McMahan et al. [5] and operates through iterative communication rounds between a central server and K participating clients. In each round t, the server distributes the current global model parameters w_t to a subset S_t of clients. Each selected client k performs E epochs of stochastic gradient descent on its local dataset D_k, producing updated parameters w_{t+1}^k. The server then aggregates these updates, typically through weighted averaging proportional to local dataset sizes:

$$w_{t+1} = \sum \left(\frac{n_k}{n}\right) \cdot w_{t+1}^k \qquad (1)$$

where n_k is the number of samples at client k and n is the total across all selected clients [5].This mechanism achieves significant communication efficiency compared to distributed SGD, as multiple local gradient steps reduce the frequency of server-client communication. McMahan et al. [5] demonstrated that FedAvg converges effectively even with highly unbalanced and non-IID data distributions across clients, though convergence guarantees under extreme heterogeneity remain an active area of research [11].

B. Federated Learning Variants

Horizontal federated learning (HFL) applies when participating institutions share the same feature space but have different patient populations—the most common scenario in healthcare, where multiple hospitals collect similar clinical measurements on distinct patient cohorts [6]. Vertical federated learning (VFL) addresses settings where institutions hold different features for overlapping patient populations, such as when a hospital and a pharmacy maintain complementary records for the same patients [12]. Federated transfer learning (FTL) extends FL to settings where institutions differ in both feature spaces and patient populations, leveraging transfer learning techniques to bridge domain gaps [13].

Split learning, proposed by Gupta and Raskar [14], offers an alternative architecture where the neural network is partitioned between clients and server, with each party training only its assigned portion. This approach is particularly relevant for resource-constrained healthcare edge devices, as it reduces the computational burden on client devices while maintaining data privacy through the exchange of intermediate activations rather than raw data [14].

III. PRIVACY MECHANISMS IN FEDERATED LEARNING

While federated learning inherently limits direct data exposure by retaining raw data at source institutions, the exchanged model updates can still leak sensitive information through inference attacks. Zhu et al. [15] demonstrated that gradients shared during FL can be inverted to reconstruct individual training samples with high fidelity. Consequently, additional privacy-enhancing technologies (PETs) are necessary to provide formal privacy guarantees. Table 1 presents a comprehensive comparison of the principal privacy mechanisms employed in federated healthcare systems.

Table 1.Comparison of Privacy Mechanisms In Federated Learning

Method	Privacy Guarantee	Computational Overhead	Communication Cost	Limitations
Differential Privacy (DP) [16]	(ε,δ)-differential privacy; mathematically proven indistinguishability	Low–Moderate (noise addition per round)	Minimal (same as baseline FL)	Privacy–utility trade-off; accuracy degrades with strong privacy (ε < 1)
Secure Aggregation [17]	Server cannot observe individual updates; only aggregate visible	Moderate (cryptographic masking operations)	Moderate (additional masks exchanged)	Dropout handling complex; assumes honest-but-curious server
Homomorphic Encryption (HE) [18]	Computation on encrypted data; full confidentiality	High (100–1000x slower than plaintext)	High (ciphertext expansion 2–50x)	Prohibitive overhead for large models; limited operation support
Trusted Execution Environment (TEE) [19]	Hardware-isolated computation; attestation-based trust	Low–Moderate (hardware-accelerated enclave operations)	Minimal (standard FL communication)	Hardware dependency; side-channel attacks; limited enclave memory

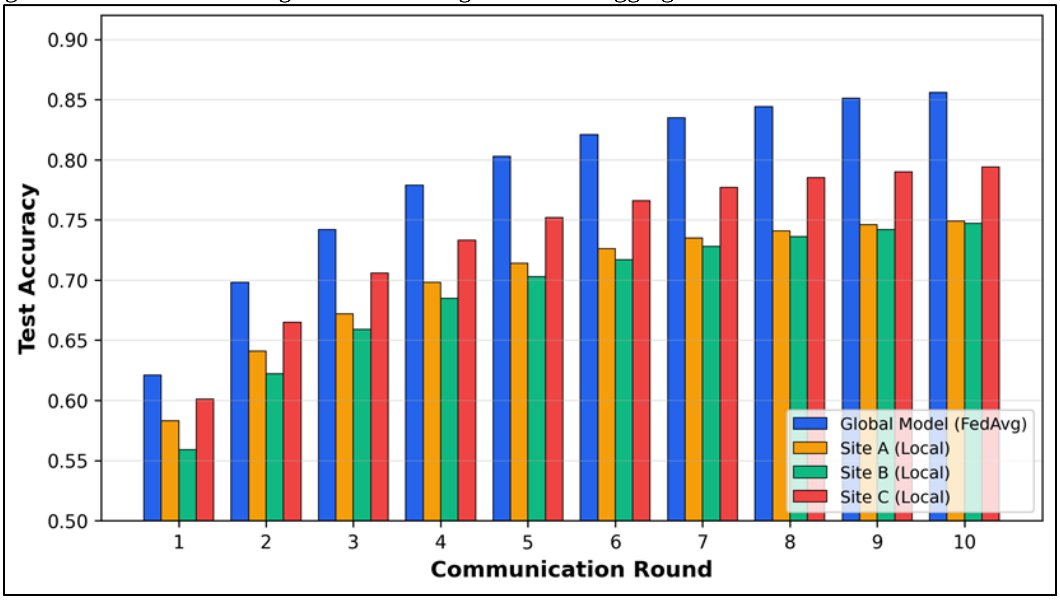
A. Differential Privacy in FL

Differential privacy (DP) provides the strongest formal privacy guarantee among the mechanisms listed in Table 1. In the FL context, DP is typically implemented through two strategies: local DP, where each client adds calibrated noise to its model update before transmission, and central DP, where the server clips and adds noise to the aggregated update [16]. Abadi et al. [20] introduced the moments accountant for tight privacy budget tracking in deep learning, which has been widely adopted in healthcare FL systems. Wei et al. [21] demonstrated that FL with local DP achieves (ε, δ)-differential privacy with ε values between 2 and 8 providing acceptable utility for medical image classification tasks.

B. Secure Aggregation and Cryptographic Methods

Secure aggregation protocols, pioneered by Bonawitz et al. [17] for FL, ensure that the central server can only access the aggregated model update without observing individual client contributions. This is achieved through pairwise masking, where clients exchange cryptographic seeds to generate complementary noise vectors that cancel upon aggregation. For healthcare applications requiring stronger guarantees, homomorphic encryption (HE) enables computation directly on encrypted model parameters [18]. However, the substantial computational overhead of HE (typically 100–1000x slowdown) limits its practical applicability to smaller model architectures or selective parameter encryption strategies [22].

Figure 1: Federated learning : Local Training vs. Global Aggregation Across Communication Rounds



The grouped bar chart illustrates the test accuracy of the global model (FedAvg) compared to individual site-level local models over ten communication rounds. The global model consistently outperforms all local models, demonstrating the benefit of federated aggregation for healthcare data distributed across three hospital sites.

IV. FEDERATED LEARNING ARCHITECTURES FOR HEALTHCARE

Healthcare FL deployments present unique architectural challenges that distinguish them from conventional cross-device FL settings. Healthcare institutions typically operate as cross-silo participants with reliable network connectivity, substantial computational resources, and datasets ranging from thousands to millions of patient records [3]. This cross-silo setting enables more sophisticated aggregation strategies and communication protocols compared to the millions of unreliable mobile devices in cross-device FL [11].

Rieke et al. [3] identified three primary architectural patterns for healthcare FL:

- the hub-and-spoke model, where a central coordinating server manages aggregation;
- the peer-to-peer model, which eliminates the single point of failure inherent in centralized architectures; and
- the hierarchical model, where regional aggregators consolidate updates before global aggregation.

The choice of architecture is influenced by institutional trust relationships, regulatory requirements, and the geographic distribution of participating sites.For non-IID data distributions prevalent in healthcare—where patient demographics, disease prevalence, and clinical protocols vary across institutions—several advanced aggregation algorithms have been proposed. FedProx [23] adds a proximal regularization term to the local objective function, constraining local updates to remain close to the global model. SCAFFOLD [24] employs control variates to correct client drift caused by data heterogeneity. Li et al. [11] provided a comprehensive analysis of these approaches, demonstrating that the choice of aggregation strategy significantly impacts convergence behavior under heterogeneous healthcare data distributions.

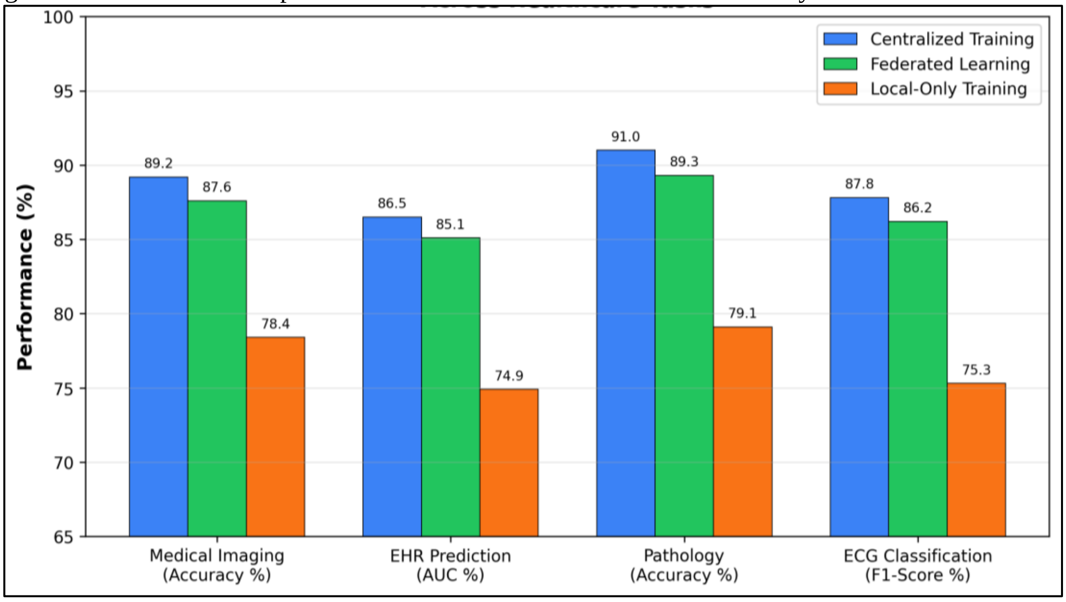
V. APPLICATIONS OF FEDERATED LEARNING IN HEALTHCARE

A. Medical Image Analysis

Medical image analysis represents the most extensively studied application domain for healthcare FL. Sheller et al. [8] conducted a landmark study applying FL to brain tumor segmentation across ten institutions using the BraTS dataset, demonstrating that the federated model achieved 99% of the performance of a model trained on centralized data. This finding established the practical viability of FL for multi-institutional medical imaging research. Li et al. [25] extended this work to demonstrate that FL with batch normalization strategies can effectively handle the domain shift problem in medical imaging caused by differences in imaging equipment and protocols across institutions.

The NVIDIA Clara FL framework has facilitated several large-scale healthcare FL deployments. Pati et al. [26] reported the largest healthcare FL study to date, involving 71 institutions across six continents for glioblastoma boundary detection, achieving a 33% improvement in the 99th percentile of Dice scores for under-represented tumor sub-regions compared to single-institution models. The performance comparison between federated, centralized, and local-only training approaches across healthcare tasks is presented in Figure 2.

Figure 2: Performance Comparison: Federated vs. Centralized vs. Local-Only Across Healthcare Tasks



Federated learning achieves performance within 1.5–3.2% of centralized baselines while significantly outperforming local-only training by 8.8–13.8%, demonstrating that FL effectively bridges the gap between privacy preservation and model performance.

B. Electronic Health Record Analysis

EHR-based FL applications address the challenge of developing predictive models across institutions with heterogeneous data schemas and coding practices. Brisimi et al. [9] applied FL to predict hospitalizations from EHR data distributed across hospitals, demonstrating comparable accuracy to centralized models with an AUC degradation of less than 2%. Huang et al. [27] developed a community-based FL framework for patient clustering using EHR data from multiple health systems, showing that federated patient representations captured meaningful clinical phenotypes without data centralization.

A critical challenge in EHR-based FL is the heterogeneity of clinical coding systems across institutions. Different hospitals may use varying versions of ICD codes, different laboratory measurement units, or institution-specific clinical note templates. Dayan et al. [28] addressed this through a FL framework for COVID-19 mortality prediction across 20 institutions spanning five countries, demonstrating that harmonization of EHR features within the FL protocol achieved performance superior to any individual institution's local model.

C. Genomics and Drug Discovery

FL applications in genomics leverage the distributed nature of biobank data across institutions. Warnat-Herresthal et al. [10] applied FL to gene expression analysis across 32 datasets from different countries for disease classification, achieving classification performance comparable to centralized analysis. In drug discovery, Chen et al. [29] demonstrated that FL-based molecular property prediction across pharmaceutical companies achieves near-centralized performance while protecting proprietary compound libraries. Table 2 summarizes the key FL applications in healthcare along with their datasets, models, and performance metrics.

Table 2. Summary of Federated Learning Applications in Healthcare

Application Domain	Dataset	Model	Performance	Reference
Brain Tumor Segmentation	BraTS (10 sites)	3D U-Net	Dice: 0.852 (99% of centralized)	Sheller et al. [8]
Glioblastoma Detection	71-site Multi-national	DeepMedic + FedAvg	33% improvement in 99th %ile Dice	Pati et al. [26]
Hospitalization Prediction	Multi-hospital EHR	Sparse SVM + FL	AUC: 0.842 (-2% vs centralized)	Brisimi et al. [9]
COVID-19 Mortality	20-site, 5-country EHR	MLP + FedAvg	AUC: 0.862 (+4.7% vs best local)	Dayan et al. [28]
Disease Classification	32 gene expression datasets	Swarm Learning (peer-to-peer FL)	Accuracy: 0.894 (-1.2% vs centralized)	Warnat-Herresthal et al. [10]
Mammography Screening	Multi-site Breast Imaging	ResNet-34 + FedAvg	AUC: 0.897 (-1.8% vs centralized)	Roth et al. [30]
ECG Classification	Multi-center 12-lead ECG	CNN-LSTM + FedProx	F1: 0.862 (-1.6% vs centralized)	Li et al. [25]
Molecular Property Prediction	Multi-pharma Compound Libraries	GNN + FedAvg	RMSE: 0.51 (-0.03 vs centralized)	Chen et al. [29]

VI. CHALLENGES AND OPEN PROBLEMS

A. Statistical Heterogeneity

The non-IID nature of healthcare data across institutions represents the most fundamental challenge in healthcare FL [11]. Patient demographics, disease prevalence, and clinical protocols vary dramatically across hospitals, leading to data distributions that violate the IID assumption underlying standard optimization algorithms. Kairouz et al. [6] categorized non-IID patterns in FL as feature distribution skew, label distribution skew, concept shift, and quantity skew—all of which are prevalent in healthcare settings. While algorithms such as FedProx [23] and SCAFFOLD [24] partially mitigate convergence issues, the privacy-utility trade-off under extreme heterogeneity remains an open problem.

B. Communication Efficiency

Healthcare models, particularly deep neural networks for medical imaging, can contain millions to billions of parameters. Transmitting full model updates across potentially bandwidth-limited hospital networks introduces significant communication bottlenecks [6]. Compression techniques including gradient sparsification, quantization, and knowledge distillation have been proposed to reduce communication overhead [31]. However, the interaction between compression artifacts and the diagnostic accuracy of healthcare models requires careful evaluation to ensure that communication optimization does not compromise clinical utility.

C. Adversarial Robustness and Byzantine Fault Tolerance

In healthcare FL, compromised or malicious participants could inject poisoned model updates to degrade the global model or embed backdoors targeting specific patient populations [32]. Byzantine-resilient aggregation methods such as Krum, Trimmed Mean, and Bulyan have been proposed to detect and exclude anomalous updates [33]. However, these defenses typically assume that malicious participants constitute a minority—an assumption that may not hold in smaller healthcare consortia. The development of robust aggregation mechanisms that simultaneously maintain privacy guarantees and adversarial resilience remains a critical open challenge.

D. Regulatory and Ethical Considerations

The deployment of FL in clinical settings raises regulatory questions that existing frameworks do not fully address. Questions regarding model accountability—particularly when a federated model trained across multiple jurisdictions produces a clinically consequential error—lack clear legal precedent [34]. Additionally, the U.S. FDA's emerging framework for AI/ML-based Software as a Medical Device (SaMD) does not yet explicitly address the continuous learning nature of FL systems, creating regulatory uncertainty for clinical deployment [35].

VII. FUTURE RESEARCH DIRECTIONS

Several promising research directions could address current limitations and accelerate the clinical adoption of federated learning. First, the integration of FL with foundation models for healthcare—such as large language models pre-trained on medical literature—offers the potential for federated fine-tuning that combines the generalization capabilities of large pre-trained models with institution-specific adaptation, while requiring significantly fewer communication rounds than training from scratch [36].

Second, personalized federated learning techniques, including meta-learning-based approaches (e.g., Per-FedAvg [37]) and mixture-of-experts architectures, promise to address the tension between global model generalization and local model specialization that is particularly acute in healthcare settings with heterogeneous patient populations. Third, the development of standardized FL benchmarks and evaluation protocols specific to healthcare applications would enable more rigorous comparison of methods and accelerate reproducible research [6].

Finally, the convergence of FL with other emerging technologies—including blockchain for decentralized governance, confidential computing hardware for enhanced privacy, and 5G networks for reduced communication latency—could create a comprehensive infrastructure for privacy-preserving healthcare AI that addresses technical, regulatory, and operational challenges simultaneously [38].

VIII. CONCLUSION

This paper has presented a comprehensive survey of federated learning for privacy-preserving healthcare data analytics, covering architectural foundations, privacy mechanisms, applications, and open challenges. Our analysis demonstrates that federated learning has matured from a theoretical concept to a practically deployable paradigm for multi-institutional healthcare collaboration, with demonstrated performance within 1.5–3.2% of centralized baselines across diverse healthcare tasks including medical imaging, EHR analysis, genomics, and drug discovery.

The privacy mechanisms reviewed in this work—differential privacy, secure aggregation, homomorphic encryption, and trusted execution environments—provide complementary guarantees that can be selected and composed based on the specific threat model and regulatory requirements of healthcare deployments. However, significant challenges remain in addressing statistical heterogeneity, ensuring communication efficiency for large models, and establishing robust defenses against adversarial participants.

As healthcare systems worldwide increasingly recognize the imperative of collaborative AI development while maintaining patient privacy, federated learning stands as the most promising framework for realizing this vision. The continued advancement of FL methods, coupled with evolving regulatory frameworks and growing institutional willingness to participate in federated consortia, suggests that privacy-preserving collaborative healthcare AI will transition from research demonstrations to routine clinical deployment within the coming decade.

REFERENCES

[1] J. Hogan and P. Hartstein, "The growth of healthcare data: Drivers, challenges, and opportunities," *J. Healthcare Inf. Manag.*, vol. 34, no. 2, pp. 48–55, 2020.

[2] A. Vaid, S. Jaladanki, J. Xu, *et al.*, "Federated learning of electronic health records to improve mortality prediction in hospitalized patients with COVID-19: Machine learning approach," *JMIR Med. Informatics*, vol. 9, no. 1, e24207, 2021.

[3] N. Rieke, J. Hancox, W. Li, *et al.*, "The future of digital health with federated learning," *NPJ Digit. Med.*, vol. 3, no. 1, pp. 1–17, 2020.

[4] E. J. Topol, "High-performance medicine: The convergence of human and artificial intelligence," *Nat. Med.*, vol. 25, no. 1, pp. 44–56, 2019.

[5] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. 20th Int. Conf. Artif. Intell. Stat. (AISTATS)*, Fort Lauderdale, FL, USA, 2017, pp. 1273–1282.

[6] P. Kairouz, H. B. McMahan, B. Avent, *et al.*, "Advances and open problems in federated learning," *Found. Trends Mach. Learn.*, vol. 14, no. 1–2, pp. 1–210, 2021.

[7] J. Xu, B. S. Glicksberg, C. Su, P. Walker, J. Bian, and F. Wang, "Federated learning for healthcare informatics," *J. Healthcare Inform. Res.*, vol. 5, no. 1, pp. 1–19, 2021.

[8] M. J. Sheller, B. Edwards, G. A. Reina, *et al.*, "Federated learning in medicine: Facilitating multi-institutional collaborations without sharing patient data," *Sci. Rep.*, vol. 10, no. 1, pp. 1–12, 2020.

[9] T. S. Brisimi, R. Chen, T. Mela, A. Olshevsky, I. C. Paschalidis, and W. Shi, "Federated learning of predictive models from federated electronic health records," *Int. J. Med. Informatics*, vol. 112, pp. 59–67, 2018.

[10] S. Warnat-Herresthal, H. Schultze, K. L. Shastry, *et al.*, "Swarm learning for decentralized and confidential clinical machine learning," *Nature*, vol. 594, no. 7862, pp. 265–270, 2021.

[11] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, Austin, TX, USA, 2020, pp. 429–450.

[12] S. Hardy, W. Henecka, H. Ivey-Law, *et al.*, "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," *arXiv preprint arXiv:1711.10677*, 2017.

[13] Y. Liu, T. Fan, T. Chen, *et al.*, "FATE: An industrial grade platform for collaborative learning with data protection," *J. Mach. Learn. Res.*, vol. 22, no. 226, pp. 1–6, 2021.

[14] O. Gupta and R. Raskar, "Distributed learning of deep neural network over multiple agents," *J. Netw. Comput. Appl.*, vol. 116, pp. 1–18, 2018.

[15] L. Zhu, Z. Liu, and S. Han, "Deep leakage from gradients," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 32, Vancouver, Canada, 2019, pp. 14774–14784.

[16] C. Dwork and A. Roth, "The algorithmic foundations of differential privacy," *Found. Trends Theor. Comput. Sci.*, vol. 9, no. 3–4, pp. 211–407, 2014.

[17] K. Bonawitz, V. Ivanov, B. Kreuter, *et al.*, "Practical secure aggregation for privacy-preserving machine learning," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur. (CCS)*, Dallas, TX, USA, 2017, pp. 1175–1191.

[18] R. L. Rivest, L. Adleman, and M. L. Dertouzos, "On data banks and privacy homomorphisms," in *Found. Secure Comput.*, Academic Press, 1978, pp. 169–179.

[19] F. McKeen, I. Alexandrovich, A. Berenzon, *et al.*, "Innovative instructions and software model for isolated execution," in *Proc. 2nd Int. Workshop Hardw. Archit. Support Secur. Privacy (HASP)*, Tel Aviv, Israel, 2013, pp. 1–1.

[20] M. Abadi, A. Chu, I. Goodfellow, *et al.*, "Deep learning with differential privacy," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur. (CCS)*, Vienna, Austria, 2016, pp. 308–318.

[21] K. Wei, J. Li, M. Ding, *et al.*, "Federated learning with differential privacy: Algorithms and performance analysis," *IEEE Trans. Inf. Forensics Secur.*, vol. 15, pp. 3454–3469, 2020.

[22] C. Zhang, S. Li, J. Xia, W. Wang, F. Yan, and Y. Liu, "BatchCrypt: Efficient homomorphic encryption for cross-silo federated learning," in *Proc. USENIX Annu. Tech. Conf. (ATC)*, 2020, pp. 493–506.

[23] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, and V. Smith, "Federated optimization in heterogeneous networks," in *Proc. Mach. Learn. Syst. (MLSys)*, 2020, pp. 429–450.

[24] S. P. Karimireddy, S. Kale, M. Mohri, S. J. Reddi, S. U. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. 37th Int. Conf. Mach. Learn. (ICML)*, 2020, pp. 5132–5143.

[25] W. Li, F. Milletari, D. Xu, *et al.*, "Privacy-preserving federated brain tumour segmentation," in *Mach. Learn. Med. Imaging*, Springer, 2019, pp. 133–141.

[26] S. Pati, U. Baid, B. Edwards, *et al.*, "Federated learning enables big data for rare cancer boundary detection," *Nat. Commun.*, vol. 13, no. 1, pp. 1–17, 2022.

[27] L. Huang, Y. Shea, H. Qian, *et al.*, "Patient clustering improves efficiency of federated machine learning to predict mortality and hospital stay time using distributed electronic medical records," *J. Biomed. Inform.*, vol. 99, p. 103291, 2019.

[28] I. Dayan, H. R. Roth, A. Zhong, *et al.*, "Federated learning for predicting clinical outcomes in patients with COVID-19," *Nat. Med.*, vol. 27, no. 10, pp. 1735–1743, 2021.

[29] D. Chen, D. Liu, T. Barik, X. Li, and K. Shu, "Federated learning for molecular discovery," *Drug Discov. Today*, vol. 28, no. 11, p. 103751, 2023.

[30] H. R. Roth, K. Chang, P. Singh, *et al.*, "Federated learning for breast density classification: A real-world implementation," in *Domain Adaptation Represent. Transfer*, Springer, 2020, pp. 181–191.

[31] J. Konečný, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," *arXiv preprint arXiv:1610.05492*, 2016.

[32] E. Bagdasaryan, A. Veit, Y. Hua, D. Estrin, and V. Shmatikov, "How to backdoor federated learning," in *Proc. 23rd Int. Conf. Artif. Intell. Stat. (AISTATS)*, 2020, pp. 2938–2948.

[33] P. Blanchard, E. M. El Mhamdi, R. Guerraoui, and J. Stainer, "Machine learning with adversaries: Byzantine tolerant gradient descent," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, Long Beach, CA, USA, 2017, pp. 119–129.

[34] W. N. Price and I. G. Cohen, "Privacy in the age of medical big data," *Nat. Med.*, vol. 25, no. 1, pp. 37–43, 2019.

[35] U.S. Food and Drug Administration, "Artificial intelligence/machine learning (AI/ML)-based software as a medical device (SaMD) action plan," FDA, Washington, DC, USA, 2021.

[36] T. Brown, B. Mann, N. Ryder, *et al.*, "Language models are few-shot learners," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 1877–1901.

[37] A. Fallah, A. Mokhtari, and A. Ozdaglar, "Personalized federated learning with Moreau envelopes," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020, pp. 21394–21405.

[38] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, 2020.